

LECTURE-2

Example: Obtain the differential equation of the family of plane curves represented by all the circles with centres on the x-axis.

Solution: Equation of the circles with centres on the x-axis

$$(x - h)^2 + y^2 = r^2,$$

where $(h, 0)$ is the centre and r is the radius of the circle. Differentiating both sides with respect to x gives us

$$2(x - h) + 2yy' = 0$$

Differentiating both sides again w.r. to x gives us

$$2(x - h) + 2yy' = 0$$

$$yy'' + y'^2 + 1 = 0$$

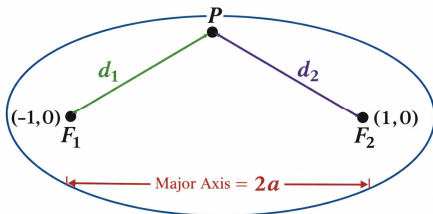
Example: Obtain the differential equation of the family of plane curves represented by all the ellipses with foci at $(-1, 0)$ and $(1, 0)$.

Solution: We know that the sum of the focal distances of a point on the ellipse is a constant and is equal to the length of its major axis, so that

$$\sqrt{(x+1)^2 + y^2} + \sqrt{(x-1)^2 + y^2} = 2a, \quad a > 1$$

for any point (x, y) on the ellipse.

The focal distances of a point on the ellipse is a constant and is equal to the length of its major axis.



$$d_1 + d_2 = 2a$$

$$\sqrt{(x+1)^2 + y^2} + \sqrt{(x-1)^2 + y^2} = 2a, \quad a > 1.$$

Then

$$\sqrt{(x+1)^2 + y^2} = 2a - \sqrt{(x-1)^2 + y^2}$$

$$(x+1)^2 + y^2 = 4a^2 + (x-1)^2 + y^2 - 4a\sqrt{(x-1)^2 + y^2}$$

$$a\sqrt{(x-1)^2 + y^2} = a^2 - x$$

$$\frac{x^2}{a^2} + \frac{y^2}{a^2 - 1} = 1, \quad a > 1$$

$$\frac{x}{a^2} + \frac{yy'}{a^2 - 1} = 0$$

$$a^2 = \frac{x}{x + yy'} \quad \& \quad a^2 - 1 = \frac{-yy'}{x + yy'}$$

$$(x + yy')(xy' - y) = y'$$

Exercises: Obtain the differential equation of the family of following curves.

1. Straight lines at a fixed distance p from the origin.

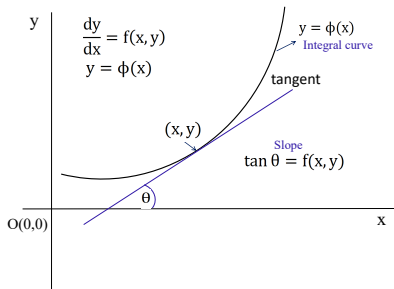
Ans.

$$\begin{aligned}x \cos \alpha + y \sin \alpha &= p \\(xy' - y)^2 &= p^2(1 + y'^2)\end{aligned}$$

2. Family of all parabolas with the focus at $(0,0)$ and with axis as the y -axis.
3. All circles.
4. All parabolas in a plane.

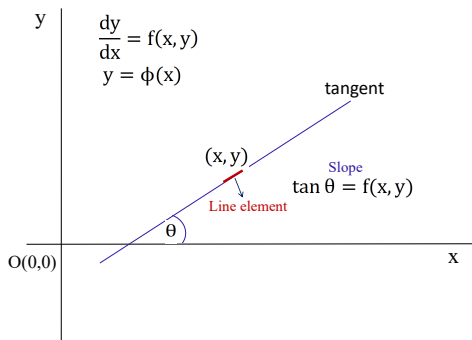
Geometric Significance

$$\frac{dy}{dx} = f(x, y) \quad (3.1)$$



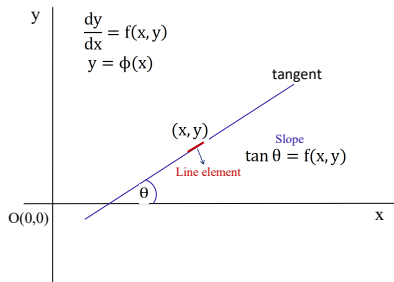
The explicit solutions of (3.1) are certain real functions, and the graphs of these solution functions are curves in the xy plane called the **integral curves** of (3.1).

The **slope** of the integral curve of (3.1) at point (x, y) is $f(x, y)$. That is the slope of the tangent of the integral curve of (3.1) at point (x, y) is $f(x, y)$.



Definition

Through the point (x, y) a short segment of the tangent to the integral curve is called a **line element**.



Definition

For each point (x, y) at which $f(x, y)$ is defined, the differential equation (1) thus defines a line segment with slope $f(x, y)$, or, in other words, a direction. Each such point, taken together with the corresponding direction so defined, constitutes the so-called **direction field** of the

differential equation (1). $\frac{dy}{dx} = 2x + y = f(x, y)$ then

$$f(1, 2) = 4 \Rightarrow \tan \theta = 4 \Rightarrow \theta \approx 76^\circ$$

Definition

Isoclines: Consider the differential equation

$$\frac{dy}{dx} = f(x, y) \quad (3.2)$$

A curve along which the slope $f(x, y)$ has a constant value c is called an isocline of the differential equation (3.2). That is, the isoclines of (3.2) are the curves $f(x, y) = c$, for different values of the parameter c .

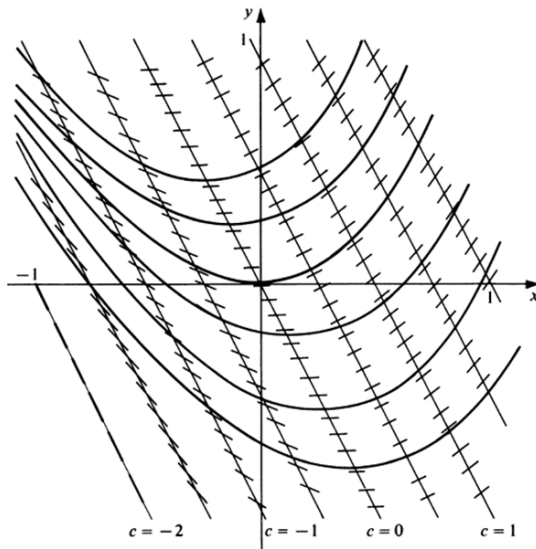
Example 1: Consider the equation

$$\frac{dy}{dx} = x.$$

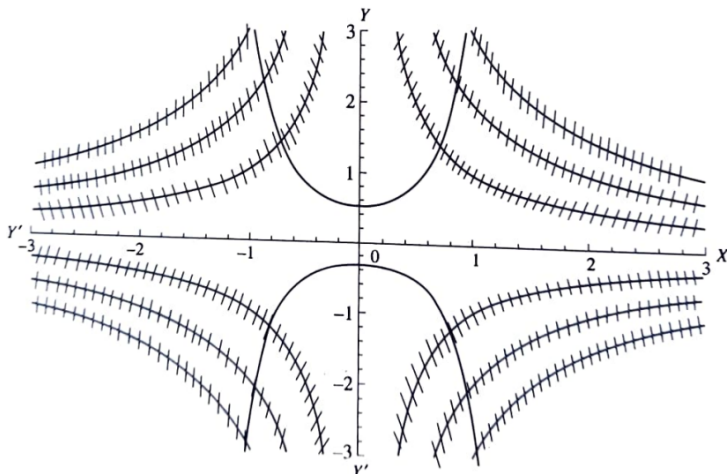
The isoclines are the curves

$$x = \lambda.$$

Example 2: $\frac{dy}{dx} = 2x + y$. Isoclines: $y = -2x + c$.



Example 3: $\frac{dy}{dx} = 4xy$. Isoclines: $4xy = \lambda$.



Definition

The expression

$$M(x, y)dx + N(x, y)dy \quad (1.1)$$

is called an exact differential in a domain D if there exists a function F of two real variables such that this expression equals to total differential $dF(x, y)$ for all $(x, y) \in D$. That is, expression (1.1) is an exact differential in D if there exists a function F such that

$$\frac{\partial F(x, y)}{\partial x} = M(x, y) \quad \text{and} \quad \frac{\partial F(x, y)}{\partial y} = N(x, y)$$

for all $(x, y) \in D$.

If $M(x, y)dx + N(x, y)dy$ is exact differential, then the differential equation

$$M(x, y)dx + N(x, y)dy = 0$$

is called exact differential equation.

Theorem

If $M(x, y)$ and $N(x, y)$ are continuous and have continuous first order partial derivatives then, the equation

$$M(x, y)dx + N(x, y)dy = 0$$

is exact if and only if

$$\frac{\partial M(x, y)}{\partial y} = \frac{\partial N(x, y)}{\partial x}.$$

Example 1: Find the solution to the equation

$$3x(xy - 2)dx + (x^3 + 2y)dy = 0.$$

Sol.:

$$\begin{aligned} M(x, y) &= 3x(xy - 2) \quad \& \quad N(x, y) = x^3 + 2y \\ \Rightarrow \quad M_y &= 3x^2 \quad \& \quad N_x = 3x^2 \quad \text{[Exact]} \end{aligned}$$

$\therefore$ There exists a function $F(x, y)$ s.t.,

$$\begin{aligned} F_x &= 3x(xy - 2) \quad \& \quad F_y = x^3 + 2y \\ \Rightarrow \quad F &= x^3y - 3x^2 + \psi(y) \quad \text{[Int. w.r.t. } x; \text{ taken } y \text{ as constant]} \\ \Rightarrow \quad F_y &= x^3 + \psi'(y) = x^3 + 2y \quad \Rightarrow \quad \psi'(y) = 2y. \end{aligned}$$

Thus $\psi(y) = y^2 + \text{constant}$. Hence $x^3y - 3x^2 + y^2 = c$.

Equation Reducible to the Exact Equation

Definition

Sometimes a given differential equation

$$M(x, y)dx + N(x, y)dy = 0 \quad (1),$$

which is not exact, can be rendered exact by multiplying (1) by a factor $u(x, y)$.

The function $u(x, y)$ is called an integrating factor of (1).

Rules for finding Integrating Factor

Rule-1: By inspection

$$\text{i) } d\left(\frac{y}{x}\right) = \frac{xdy - ydx}{x^2}.$$

$$\text{ii) } d[\log(xy)] = \frac{xdy + ydx}{xy}.$$

$$\text{iii) } d\left(\tan^{-1}\left(\frac{y}{x}\right)\right) = \frac{xdy - ydx}{x^2 + y^2} \text{ \& } \\ d\left(\tan^{-1}\left(\frac{x}{y}\right)\right) = \frac{ydx - xdy}{x^2 + y^2}.$$

$$\text{iv) } d\left(-\frac{1}{xy}\right) = \frac{xdy + ydx}{x^2y^2}.$$

$$\text{v) } d\left(\frac{y^2}{x^2}\right) = \frac{2x^2ydy - 2xy^2dx}{x^4}.$$

$$\text{vi) } d\left[\frac{1}{2} \log(x^2 + y^2)\right] = \frac{xdx + ydy}{x^2 + y^2}.$$

$$\text{vii) } d\left(\frac{e^x}{y}\right) = \frac{ye^x dx - e^x dy}{y^2} \text{ \& } d\left(\frac{e^y}{x}\right) = \frac{xe^y dy - e^y dx}{x^2}.$$

Rules for finding Integrating Factor

Rule-2: $Mdx + Ndy = 0$, M and N are homogeneous functions of same degree and $Mx + Ny \neq 0$ then $\frac{1}{Mx + Ny}$ is an integrating factor.

Example 2: $(x^2y - 2xy^2)dx - (x^3 - 3x^2y)dy = 0$.
 $\frac{1}{x^2y^2}$ is an integrating factor.

$$\left(\frac{1}{y} - \frac{2}{x}\right) dx - \left(\frac{x}{y^2} - \frac{3}{y}\right) dy \Rightarrow \frac{x}{y} - 2 \log x + 3 \log y = c.$$

Rules for finding Integrating Factor

Rule-3: If $Mdx + Ndy = 0$, where $M = f_1(xy)y$ and $N = xf_2(xy)$ and $xM - Ny \neq 0$. Then $\frac{1}{xM - yN}$ is an integrating factor.

Example 3:

$$(xy \sin(xy) + \cos(xy)) ydx + (xy \sin(xy) - \cos(xy)) xdy = 0.$$

Here, $xM - yN = 2xy \cos(xy) \neq 0$ is an integrating factor of the given differential equation.

$$(xy \sin(xy) + \cos(xy)) ydx + (xy \sin(xy) - \cos(xy)) xdy = 0$$

$$\Rightarrow \frac{1}{2} \left(y \tan(xy) + \frac{1}{x} \right) dx + \frac{1}{2} \left(x \tan xy - \frac{1}{y} \right) dy = 0$$

$$\Rightarrow \log \sec(xy) + \log \frac{x}{y} = \log c \Rightarrow \frac{x}{y} \sec(xy) = c.$$

Rules for finding Integrating Factor

Rule-4:

Theorem

The differential equation $M(x, y)dx + N(x, y)dy = 0$ has an integrating factor $u(x)$, which is a function of x alone if and only if $\frac{1}{N} \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right)$ is a function of x only, say $\phi(x)$, and in this case $u(x)$ must be a constant multiple of $e^{\int \phi(x) dx}$.

Rule-5:

Theorem

The differential equation $M(x, y)dx + N(x, y)dy = 0$ has an integrating factor $v(y)$, which is a function of y alone if and only if $\frac{1}{M} \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right)$ is a function of y only, say $\psi(y)$, and in this case $v(y)$ must be a constant multiple of $e^{\int \psi(y) dy}$.

Exercise 1: $xydx - (x^2 + 2y^2)dy = 0$.

Proof of Rule-4/5:

Suppose that there exists such function $u(x)$. Then $u(x)M(x, y)dx + u(x)N(x, y)dy = 0$ is exact i.e.

$$\begin{aligned}\frac{\partial}{\partial y}(uM) &= \frac{\partial}{\partial x}(uN) \\ \Rightarrow \quad \frac{1}{N} \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right) &= \frac{1}{u(x)} \frac{du(x)}{dx},\end{aligned}$$

which shows that $\frac{1}{N} \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right)$ is a function of x alone.

Conversely, if $\frac{1}{N} \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right) = \phi(x)$, a function of x alone, then the equation $e^{\int \phi(x) dx} M(x, y) dx + e^{\int \phi(x) dx} N(x, y) dy = 0$ is exact, for

$$\frac{\partial}{\partial y} \left(e^{\int \phi(x) dx} M(x, y) \right) - \frac{\partial}{\partial x} \left(e^{\int \phi(x) dx} N(x, y) \right) = 0$$

$$i.e., \quad e^{\int \phi(x) dx} \frac{\partial M}{\partial y} - e^{\int \phi(x) dx} \phi(x) N(x, y) - e^{\int \phi(x) dx} \frac{\partial N}{\partial x} = 0$$

$$i.e., \quad e^{\int \phi(x) dx} \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right) - e^{\int \phi(x) dx} \phi(x) N(x, y) = 0$$

$$[\because \frac{1}{N} \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right) = \phi(x)].$$

Therefore, if $\frac{1}{N} \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right)$ is a function of x alone, then

$$\frac{du}{u} = \phi(x) dx$$

which gives $u(x) = ce^{\int \phi(x) dx}$, where c is a constant